

1. Display all documents in instructors collection

```
ITI> db.instructors.find();
[
  {
    _id: 6,
    firstName: 'noha',
    lastName: 'hesham',
    age: 21,
    salary: 3500,
    address: { city: 'cairo', street: 10, building: 8 },
    courses: [ 'js', 'mvc', 'signalR', 'expressjs' ]
  },
  {
    _id: 7,
    firstName: 'mona',
    lastName: 'ahmed',

```

2. Display all instructors with salaries greater than 4000 (only show firstName and salary)

```
ITI> db.instructors.find({ salary: { $gt: 4000 } }, { firstName: 1, salary: 1 })
[
  { _id: 8, firstName: 'mazen', salary: 7040 },
  { _id: 9, firstName: 'ebtesam', salary: 7500 },
  { _id: 10, firstName: 'badr', salary: 5550 },
  { _id: 3, firstName: 'mazen', salary: 7010 },
  { _id: 4, firstName: 'ebtesam', salary: 6200 }
]
```

3. Display all instructors with ages less than or equal 25.

```
ITI> db.instructors.find({ age: { $lte: 25 } });
[
  {
    _id: 6,
    firstName: 'noha',
    lastName: 'hesham',
    age: 21,
    salary: 3500,
    address: { city: 'cairo', street: 10, building: 8 },
    courses: [ 'js', 'mvc', 'signalR', 'expressjs' ]
  },
  {
    _id: 7,
    firstName: 'mona',
    lastName: 'ahmed',
    age: 21,

```

4. Display all instructors with city mansoura and street number 10 or 14 only display (firstname,address,salary).

```
ITI> db.instructors.find(
...   {
...     $and: [
...       { "address.city": "mansoura" },
...       { $or: [{ "address.street": 10 }, { "address.street": 14 }] },
...     ],
...   },
...   { firstName: 1, address: 1, salary: 1 },
... );
[
  {
    _id: 9,
    firstName: 'ebtesam',
    salary: 7500,
    address: { city: 'mansoura', street: 14, building: 3 }
  },
  {
    _id: 4,
    firstName: 'ebtesam',
    salary: 6200,
    address: { city: 'mansoura', street: 14, building: 7 }
  }
]
```

5. Display all instructors that have js and jquery in their courses(both of them not one of them).

```
ITI> db.instructors.find({ courses: { $all: ["js", "jquery"] } });
```

6. Display number of courses for each instructor and display first name with number of courses.

```
ITI> db.instructors.find().forEach((inst) => {
...   if (inst) {
...     print(inst.firstName, inst.courses.length);
...   }
... });
noha 4
mona 4
mazen 3
ebtesam 5
badr 4
mona 4
mazen 3
```

7. Write mongodb query to get all instructors that have firstName and lastName properties , sort them by firstName ascending then by lastName descending and finally display their data FullName and age.

```
ITI> db.instructors
...   .find({ firstName: { $exists: 1 }, lastName: { $exists: 1 } })
...   .sort({ firstName: 1, lastName: -1 })
...   .forEach((inst) => {
...     print("Full name: " + inst.firstName, inst.lastName + ",Age: " + inst.age);
...   });
Full name: Asaad Mansour,Age: undefined
Full name: badr ahmed,Age: 22
Full name: ebtesam hesham,Age: 21
Full name: ebtesam ahmed,Age: 28
Full name: mazen mohammed,Age: 21
Full name: mazen ali,Age: 30
Full name: mona mohammed,Age: 21
Full name: mona ahmed,Age: 21
Full name: noha hesham,Age: 21
```

8. Delete instructor with first name “ebtesam” and has only 5 courses in courses array

```
ITI> db.instructors.deleteOne({
...   firstName: "ebtesam",
...   courses: { $size: 5 }
... });
{ acknowledged: true, deletedCount: 1 }
```

9. Add active property to all instructors and set its value to true.

```
ITI> db.instructors.updateMany({}, { $set: { active: true } });
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 9,
  modifiedCount: 9,
  upsertedCount: 0
}
```

10. Change “EF” course to “jquery” for “mazen mohammed” instructor (without knowing EF Index in courses array)

```
ITI> db.instructors.updateOne(
...   {
...     firstName: "mazen",
...     lastName: "mohammed",
...     courses: "EF",
...   },
...   {
...     $set: { "courses.$": "jquery" },
...   },
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

11. Add jquery course for "noha hesham".

```
ITI> db.instructors.updateOne(  
...   { firstName: "noha", lastName: "hesham" },  
...   { $addToSet: { courses: "jquery" } },  
... );  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

12. Remove courses property for "ahmed mohammed " instructor

```
ITI> db.instructors.updateOne(  
...   { firstName: "ahmed", lastName: "mohammed" },  
...   { $unset: { courses: "" } },  
... );  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 0,  
  modifiedCount: 0,  
  upsertedCount: 0  
}
```

13. Decrease salary by 500 for all instructors that has only 3 courses in their list (\$inc)

```
ITI> db.instructors.updateMany(  
...   { courses: { $size: 3 } },  
...   { $inc: { salary: -500 } },  
... );  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 2,  
  modifiedCount: 2,  
  upsertedCount: 0  
}
```

14. Rename address field for all instructors to fullAddress.

```
ITI> db.instructors.updateMany({}, { $rename: { address: "fullAddress" } });  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 9,  
  modifiedCount: 7,  
  upsertedCount: 0  
}
```

15. Change street number for noha hesham to 20

```
ITI> db.instructors.updateOne(
...   { firstName: "noha", lastName: "hesham" },
...   { $set: { "address.street": 20 } },
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```